

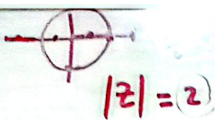
Find the residues of

$$1) f(z) = \frac{z^2}{(z-1)^2(z+2)}$$

Residue at $z = -2$:
 $\text{Res}(f(z)) = \lim_{z \rightarrow -2} (z+2) \cdot \frac{z^2}{(z-1)^2(z+2)}$
 $= \lim_{z \rightarrow -2} \frac{z^2}{(z-1)^2} = \frac{4}{9}$

Residue at $z = 1$:
 $\text{Res}(f(z)) = \lim_{z \rightarrow 1} \frac{d}{dz} \left((z-1)^2 \cdot \frac{z^2}{(z-1)^2(z+2)} \right)$
 $= \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{z^2}{z+2} \right)$
 $= \lim_{z \rightarrow 1} \left[\frac{(2z)(z+2) - z^2}{(z+2)^2} \right]$
 $= \frac{5}{9}$

$$2) f(z) = \frac{\sin z}{z \cos(z)}$$



Residue at $z = 0$:
 $\text{Res}(f(z)) = \lim_{z \rightarrow 0} \left[z \cdot \frac{\sin(z)}{z \cos(z)} \right]$
 $= 0$

Residue at $z = \frac{\pi}{2}$:
 $\text{Res}(f(z)) = \lim_{z \rightarrow \frac{\pi}{2}} \left((z - \frac{\pi}{2}) \cdot \frac{\sin(z)}{z \cos(z)} \right)$
 $= \lim_{z \rightarrow \frac{\pi}{2}} \frac{\sin(z)}{z} \cdot \lim_{z \rightarrow \frac{\pi}{2}} \frac{z - \frac{\pi}{2}}{\cos(z)}$
 $= \frac{1}{\frac{\pi}{2}} \cdot \lim_{z \rightarrow \frac{\pi}{2}} \frac{1}{-\sin(z)} = \frac{2}{\pi} \left(\frac{1}{-1} \right) = -\frac{2}{\pi}$

Residue at $z = -\frac{\pi}{2}$:
 $\text{Res}(f(z)) = \lim_{z \rightarrow -\frac{\pi}{2}} \left((z + \frac{\pi}{2}) \cdot \frac{\sin(z)}{z \cos(z)} \right)$
 $= \dots$

$$3) f(z) = \frac{\cos(z)}{z \sin(z)}$$

$$|z| = 3$$

$z = 0$ | $\sin(z) = 0$
 $z = 0, \pi, -\pi$

Residue at $z = 0$:
 $\text{Res}(f(z)) = \lim_{z \rightarrow 0} \left[\frac{1}{1!} \lim_{z \rightarrow 0} \frac{d}{dz} \left(z \cdot \frac{\cos(z)}{z \sin(z)} \right) \right]$

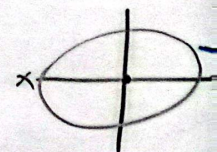
$$= \lim_{z \rightarrow 0} \frac{d}{dz} \left[z \cdot \frac{\cos(z)}{\sin(z)} \right] = \lim_{z \rightarrow 0} \frac{d}{dz} [z \cot(z)]$$

$$= \lim_{z \rightarrow 0} \left[(1) \cot(z) + (-\csc^2(z)) z \right]$$

$$= \lim_{z \rightarrow 0} \left[\frac{\cos(z)}{\sin(z)} - \frac{z}{\sin^2(z)} \right] = \lim_{z \rightarrow 0} \frac{\sin(z) \cos(z) - z}{\sin^2(z)}$$

$$= \lim_{z \rightarrow 0} \frac{\frac{1}{2} \sin(2z) - z}{\sin^2(z)} = \lim_{z \rightarrow 0} \frac{\frac{1}{2} \cos(2z)(2) - 1}{2 \sin(z) \cos(z)} \quad (\sin z)$$

$$= \lim_{z \rightarrow 0} \frac{\cos(2z) - 1}{2 \sin(z) \cos(z)} = \lim_{z \rightarrow 0} \frac{-2 \sin^2(z)}{2 \sin(z) \cos(z)} = \frac{0}{1} = 0$$



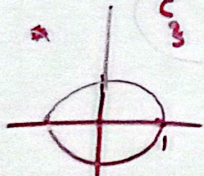
$$\oint_C \frac{z-3}{z^2+2z+5} dz$$

$$C: |z|=1$$

$$C: |z+1-i|=2$$

$$C: |z+1+i|=2$$

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sol

$$|z|=1$$

$z-3$ analytic inside and on $C: |z|=1$

$$z = -1 + 2i$$

$$z^2 + 2z + 5 = 0$$

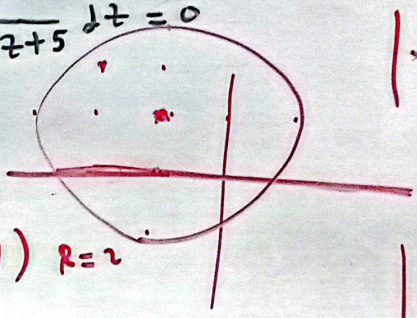
$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \begin{matrix} a=1 \\ b=2 \\ c=5 \end{matrix}$$

$$z = -1 - 2i$$

$$z = \frac{-2 \pm \sqrt{4 - 4(1)(5)}}{2(1)} \quad \begin{matrix} 4-20 \\ = -16 \end{matrix}$$

$$z = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$$

$$1) |z|=1 \quad \int \frac{z-3}{z^2+2z+5} dz = 0$$



$$|-1+2i + 1-i| = |i| = 1$$

$$2) |z+1-i|=2$$

$$(-1, 1) \quad R=2$$

$$|-1-2i + 1-i| = |-3i| = 3$$

$z = -1 + 2i$ inside C

$z = -1 - 2i$ out side C

$$\int \frac{z-3}{(z+1-2i)(z+1+2i)} dz$$

$$= \int \frac{z-3}{z+1+2i} dz = 2\pi i \left[\frac{z-3}{z+1+2i} \right]_{z=-1+2i}$$

$$= 2\pi i \left[\frac{-1+2i-3}{-1+2i+1+2i} \right] = 2\pi i \left[\frac{-4+2i}{4i} \right]$$

$$\begin{aligned} &= \pi \left(\frac{-4+2i}{2} \right) \\ &= \pi [-2+i] \\ &= \pi \left(\frac{-4+2i}{4i} \right) \end{aligned}$$

$$\frac{1}{1+z} = 1 - z + z^2 - z^3 + z^4 - \dots \quad (|z| < 1)$$

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots$$

$$\frac{1}{1+z^2} = 1 - (z^2) + (z^2)^2 - (z^2)^3 + \dots$$

$$= 1 - z^2 + z^4 - z^6 + \dots$$

$$\frac{1}{3+z} = \frac{1}{3 \left(1 + \frac{z}{3}\right)}$$

$$= \frac{1}{3} \left[1 - \left(\frac{z}{3}\right) + \left(\frac{z}{3}\right)^2 - \left(\frac{z}{3}\right)^3 \dots \right]$$

$$\frac{1}{1+z} = \frac{1}{z \left(1 + \frac{1}{z}\right)} = \frac{1}{z} \left[1 - \left(\frac{1}{z}\right) + \left(\frac{1}{z}\right)^2 - \left(\frac{1}{z}\right)^3 \dots \right] \quad |z| > 1$$

$$\frac{1}{1-z} = \frac{1}{-z \left(1 - \frac{1}{z}\right)} = \frac{1}{-z} \left[1 + \left(\frac{1}{z}\right) + \left(\frac{1}{z}\right)^2 + \left(\frac{1}{z}\right)^3 \dots \right] \quad \left|\frac{1}{z}\right| < 1$$

$$\frac{1}{3+z} = \frac{1}{z \left(1 + \frac{3}{z}\right)} = \frac{1}{z} \left[1 - \left(\frac{3}{z}\right) + \left(\frac{3}{z}\right)^2 - \left(\frac{3}{z}\right)^3 \dots \right] \quad |z| > 3$$

$$\frac{1}{z} = \frac{1}{z} \quad |z| < 1$$

$$\frac{1}{1+z} = 1 - z + z^2 - z^3 + z^4 - \dots \quad (|z| < 1)$$

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots$$

$$\frac{1}{(1-z)^2} = 1 + 2z + 3z^2 + \dots$$

$$\frac{1}{z-1} = \frac{1}{z-1}$$

$$|z-1| < 1$$

$$\frac{1}{z-1}$$

$$|z+2| < 1$$

$$\frac{1}{z+2-1-2} = \frac{1}{(z+2)-3}$$

$$= \frac{1}{-3 \left(1 - \frac{z+2}{3} \right)} = \frac{1}{-3} \left[1 - \left(\frac{z+2}{3} \right) + \left(\frac{z+2}{3} \right)^2 - \dots \right]$$

$$u = z+2 \Rightarrow z = u-2$$

$$\frac{1}{z-1}$$

$$|z+2| < 1$$

$$|u| < 1$$

$$= \frac{1}{u-2-1} = \frac{1}{u-3} = \frac{1}{-3 \left(1 - \frac{u}{3} \right)}$$

$$= \frac{1}{-3} \left[1 - \left(\frac{u}{3} \right) + \left(\frac{u}{3} \right)^2 - \dots \right]$$

$$= \frac{1}{-3} \left[- \left(\frac{z+2}{3} \right) + \left(\frac{z+2}{3} \right)^2 - \dots \right]$$

$$\int \frac{z}{z} dz$$

تفصيل

$$x = a(\theta + \sin \theta)$$

$$y = a(1 - \cos(\theta))$$

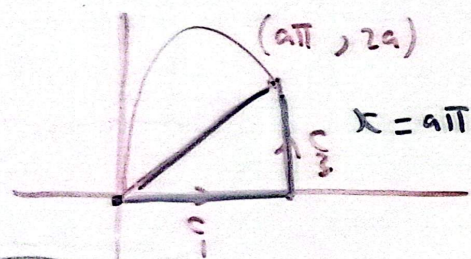
$$z = x + iy$$

$$(0, 0) \quad (a\pi, 2a)$$

$$\int_{\gamma_1} + \int_{\gamma_2}$$

$$(y=0)$$

$$(x=a\pi)$$



$$z + i$$

$$\int_0^1 (\bar{z})^2 dz$$

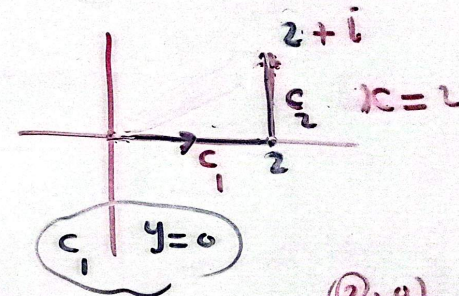
ليس تغير

c the real ax is 2 and the vertically to $z + i$

$$\int_{\gamma_1} (y=0)$$

$$+ \int_{\gamma_2}$$

$$(x=2)$$



$$(2, 0)$$

$$(2, 1)$$

$$\frac{y-0}{x-2} = \frac{1-0}{2-2}$$

$$\frac{y}{x-2} \times \frac{1}{0}$$

$$\int_C \frac{z^2}{(z-1)^2(z+2)} dz \quad C: |z|=2.5$$

$$\frac{z^2}{(z-1)^2(z+2)} = \frac{A}{z-1} + \frac{B}{(z-1)^2} + \frac{C}{z+2}$$

$$z^2 = A(z-1)(z+2) + B(z+2) + C(z-1)^2$$

$$z=1 \quad 1 = 3B \Rightarrow B = \frac{1}{3} = \frac{3}{9}$$

$$z=-2 \quad 4 = C(9) \Rightarrow C = \frac{4}{9}$$

$$z=0 \quad 0 = A(-1)(2) + \left(\frac{3}{9}\right)(2) + \frac{4}{9}(1)$$

$$0 = -2A + \frac{10}{9} \Rightarrow 2A = \frac{10}{9} \Rightarrow A = \frac{5}{9}$$

$$\frac{z^2}{(z-1)^2(z+2)} = \frac{\frac{5}{9}}{z-1} + \frac{\frac{3}{9}}{(z-1)^2} + \frac{\frac{4}{9}}{z+2}$$

$$(1) \quad \oint_C \frac{z^2}{(z-1)^2(z+2)} dz$$

$$= \oint_C \frac{\left(\frac{5}{9}\right)}{z-1} dz + \oint_C \frac{\left(\frac{3}{9}\right)}{(z-1)^2} dz + \oint_C \frac{\frac{4}{9}}{z+2} dz$$

$$= 2\pi i \left(\frac{5}{9}\right) + 2\pi i \frac{d}{dz} \left(\frac{\frac{3}{9}}{z-1}\right) + 2\pi i \left(\frac{4}{9}\right)$$

$$= 2\pi i \left[\frac{5}{9} + \frac{4}{9} \right] = 2\pi i \left(\frac{9}{9}\right) = 2\pi i$$

$$|z|=2.5$$

$$|1|=1$$

$$|-2|=2$$

$$\int \frac{g(z)}{z-a} dz \quad \text{analytic}$$

$$= 2\pi i g(a)$$


$$\int \frac{g(z)}{(z-a)^n} dz$$

$$= \frac{2\pi i}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} (g(z)) \bigg|_{z=a}$$


$$\int_C \frac{z^2}{(z-1)^2(z+2)} dz \quad C: |z|=2.5$$

$$\frac{z^2}{(z-1)^2(z+2)} = \frac{A}{z-1} + \frac{B}{(z-1)^2} + \frac{C}{z+2}$$

$$z^2 = A(z-1)(z+2) + B(z+2) + C(z-1)^2$$

$$z=1 \quad 1 = 3B \Rightarrow B = \frac{1}{3} = \frac{3}{9}$$

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$$0 = -2A + \frac{10}{9} \Rightarrow 2A = \frac{10}{9} \Rightarrow A = \frac{5}{9}$$

$$\frac{z^2}{(z-1)^2(z+2)} = \frac{\frac{5}{9}}{z-1} + \frac{\frac{3}{9}}{(z-1)^2} + \frac{\frac{4}{9}}{z+2}$$

[2]

$$\text{Res}(f(z)) = \frac{1}{1!} \lim_{z \rightarrow 1} \frac{d}{dz} \left[(z-1)^2 \frac{z^2}{(z-1)^2(z+2)} \right] \quad \begin{matrix} z-1=0 \\ z=1 \\ z+2=0 \\ z=-2 \end{matrix}$$

$$= \lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{z^2}{z+2} \right]$$

$$= \lim_{z \rightarrow 1} \frac{(2z)(z+2) - (1)z^2}{(z+2)^2}$$

$$= \frac{(2)(3) - (1)(1)}{(3)^2} = \frac{5}{9}$$

$$\text{Res}(f(z)) = \lim_{z \rightarrow -2} (z+2) \frac{z^2}{(z-1)^2(z+2)}$$

$$= \lim_{z \rightarrow -2} \frac{z^2}{(z-1)^2}$$

$$= \frac{(-2)^2}{(-2-1)^2} = \frac{4}{9}$$

$$\int \frac{z^2}{(z-1)^2(z+2)} dz = 2\pi i \left[\text{Res}(f(z))_{z=1} + \text{Res}(f(z))_{z=-2} \right]$$

$$= 2\pi i \left[\frac{5}{9} + \frac{4}{9} \right]$$

$$= 2\pi i \left[\frac{9}{9} \right]$$

$$= 2\pi i$$

$$\int_C \frac{z^2}{(z-1)^2(z+2)} dz \quad C: |z|=2.5$$

$$\frac{z^2}{(z-1)^2(z+2)} = \frac{\frac{5}{9}}{z-1} + \frac{\frac{3}{9}}{(z-1)^2} + \frac{\frac{4}{9}}{z+2}$$

$$z=1 \quad 0 < |z-1| < 3$$

$$u = z-1 \Rightarrow z = u+1$$

$$\frac{\frac{5}{9}}{u} + \frac{\frac{3}{9}}{u^2} + \frac{\frac{4}{9}}{u+3}$$

$$= \frac{\frac{5}{9}}{u} + \frac{\frac{3}{9}}{u^2} + \frac{4}{9} \cdot \frac{1}{3} \left(\frac{1}{1+\frac{u}{3}} \right)$$

$$0 < |z+2| < 3$$

$$= \frac{\frac{5}{9}}{u} + \frac{\frac{3}{9}}{u^2} + \frac{4}{9} \cdot \frac{1}{3} \left[1 - \left(\frac{u}{3}\right) + \left(\frac{u}{3}\right)^2 - \left(\frac{u}{3}\right)^3 + \dots \right]$$

$$= \left(\frac{\frac{5}{9}}{z-1} \right) + \frac{\frac{3}{9}}{(z-1)^2} + \frac{4}{9} \left(\frac{1}{3} \right) \left[1 - \left(\frac{z-1}{3} \right) + \left(\frac{z-1}{3} \right)^2 - \dots \right]$$

$$\int_C \frac{z^2}{(z-1)^2(z+2)} dz = 2\pi i \left[\frac{5}{9} \right] + 0 + 0 + \dots$$

$$|z+2| < 3$$

كل

$$u = z+2$$